End-to-End Supply Chain Analysis Using Python

From Operational Data to Predictive Supply Chain Insights

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Role: Data Analyst
Tools & Technologies: Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn • Jupyter Notebook

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Project Overview

This project presents an end-to-end analysis of supply chain and order fulfillment operations for a global e-commerce business. The analysis focuses on understanding shipping performance, identifying delivery bottlenecks, evaluating the relationship between delays and profitability, and developing a predictive system to identify orders at risk of late delivery.

Using Python-based data analysis, statistical techniques, visualization, and machine learning, the project transforms raw supply chain data into actionable business insights.

Project Objectives

The key objectives of this project are to:

- Analyze overall supply chain and order fulfillment performance.
- Measure scheduled vs. actual shipping performance.
- Identify major delivery bottlenecks and operational risk areas.
- Analyze delay patterns across shipping modes, regions, products, and customer segments.
- Understand the impact of shipping delays on order profitability.
- Identify factors associated with late-delivery risk.
- Build a machine learning model to predict potential late shipments.
- Translate analytical and predictive findings into practical business recommendations.

Technologies Used

Category	Tools
Programming	Python
Data Manipulation	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Statistical Analysis	SciPy / Statistical Methods

Category	Tools
Machine Learning	Scikit-learn
Environment	Jupyter Notebook
Methodology	EDA, Feature Engineering, Predictive Analytics

> **Project Focus:**
> The goal is not only to understand *what happened*, but also to determine *why it happened*, *which orders are at risk*, and *what actions can improve supply chain performance*.

Business Problem

A global e-commerce company operating across multiple regions manages end-to-end order fulfillment, including shipping and delivery, for products like sporting goods. The company is facing inconsistent delivery performance, where actual shipping times often deviate from scheduled timelines, leading to late deliveries and unpredictable order profitability.

Desired Outcome:

The goal is to analyze delivery operations, identify bottlenecks, and build a predictive system to reduce delays, optimize shipping decisions, and improve overall profitability and efficiency.

```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import matplotlib.ticker as ticker
import matplotlib.cm as cm
from warnings import filterwarnings
filterwarnings('ignore')

# Set professional color theme
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("viridis")

viridis_colors = cm.viridis(np.linspace(0, 1, 5))
primary_color = viridis_colors[0]
secondary_color = viridis_colors[1]
accent_color = viridis_colors[2]
danger_color = '#800000'
neutral_color = viridis_colors[4]
custom_palette = viridis_colors

df = pd.read_csv('DataCoSupplyChainDataset.csv', encoding='latin-1')
```

Exploratory Data Analysis (EDA)

- Load and understand the dataset structure (rows, columns, data types)
- Identify missing values and duplicates
- Data Cleaning
- Explore categorical variables and their distributions
- Calculate key metrics like order processing time and delays

```
# Overview
print('rows, cols:', df.shape)
print('\ncolumns:')
print(df.columns.tolist())
print('\nNum duplicates:', df.duplicated().sum())
print('\nMissing values (top 20):')
print(df.isna().sum().sort_values(ascending=False).head(20))
```

```
rows, cols: (180519, 53)

columns:
['Type', 'Days for shipping (real)', 'Days for shipment (scheduled)',
'Benefit per order', 'Sales per customer', 'Delivery Status',
'Late_delivery_risk', 'Category Id', 'Category Name', 'Customer City',
'Customer Country', 'Customer Email', 'Customer Fname', 'Customer Id',
'Customer Lname', 'Customer Password', 'Customer Segment', 'Customer
State', 'Customer Street', 'Customer Zipcode', 'Department Id',
'Department Name', 'Latitude', 'Longitude', 'Market', 'Order City',
'Order Country', 'Order Customer Id', 'order date (DateOrders)',
'Order Id', 'Order Item Cardprod Id', 'Order Item Discount', 'Order
Item Discount Rate', 'Order Item Id', 'Order Item Product Price',
'Order Item Profit Ratio', 'Order Item Quantity', 'Sales', 'Order Item
Total', 'Order Profit Per Order', 'Order Region', 'Order State',
'Order Status', 'Order Zipcode', 'Product Card Id', 'Product Category
Id', 'Product Description', 'Product Image', 'Product Name', 'Product
Price', 'Product Status', 'shipping date (DateOrders)', 'Shipping
Mode']

Num duplicates: 0

Missing values (top 20):
Product Description         180519
Order Zipcode               155679
Customer Lname                   8
Customer Zipcode                 3
Type                             0
Order Profit Per Order           0
Order Item Cardprod Id           0
Order Item Discount              0
Order Item Discount Rate         0
Order Item Id                    0
Order Item Product Price         0
```

```
Order Item Profit Ratio          0
Order Item Quantity              0
Sales                            0
Order Item Total                 0
Order Region                     0
order date (DateOrders)          0
Order State                      0
Order Status                     0
Product Card Id                  0
dtype: int64
```

```
# Data Cleaning
columns_to_drop = [
    'Product Description',
    'Product Image',
    'Customer Email',
    'Customer Password',
    'Customer Fname',
    'Customer Lname',
    'Customer Street',
    'Customer Zipcode',
    'Order Zipcode',
    'Longitude',
    'Latitude',
    'Order Item Cardprod Id',
    'Order Item Id',
    'Order Item Discount',
    'Order Item Discount Rate',
    'Order Item Product Price',
    'Order Item Quantity',
    'Order Item Total',
    'Category Id',
    'Department Id',
    'Order Id',
    'Order Customer Id',
    'Customer Id',
    'Product Card Id',
    'Product Category Id',
    'Benefit per order', # identical to Order Profit Per Order
    'Product Status', # have only one value,
    'Customer City',
    'Order City',
    'Order Country',
    'Order State',
    'Customer State',
    'Market']

# dropping columns that are either fully missing, redundant, or have
only one value (and thus no variance)
df = df.drop(columns=columns_to_drop)
```

```python
# removing canceled orders since they are not relevant for delivery
time analysis and may have different patterns than completed orders
df = df[df['Delivery Status'] != 'Shipping canceled']

#  Standard date conversion
for c in ['order date (DateOrders)', 'shipping date (DateOrders)']:
    df[c] = pd.to_datetime(df[c], errors='coerce', dayfirst=False)

# after data cleaning, let's check the overview again to see how the
dataset has changed
print('rows, cols:', df.shape)
print('\nMissing values (top 5):')
print(df.isna().sum().sort_values(ascending=False).head(5))
```

```
rows, cols: (172765, 20)

Missing values (top 5):
Type                          0
Days for shipping (real)      0
shipping date (DateOrders)    0
Product Price                 0
Product Name                  0
dtype: int64
```

```python
# value counts for categorical columns with low cardinality
for col in df.columns:
    if df[col].nunique() < 10:
        print(f'\n{col} value counts:')
        print(df[col].value_counts())
```

```
Type value counts:
Type
DEBIT       69295
TRANSFER    42129
PAYMENT     41725
CASH        19616
Name: count, dtype: int64

Days for shipping (real) value counts:
Days for shipping (real)
2    54205
6    27489
3    27478
4    27297
5    27003
0     4839
1     4454
Name: count, dtype: int64
```

```
Days for shipment (scheduled) value counts:
Days for shipment (scheduled)
4    103153
2     33806
1     26513
0      9293
Name: count, dtype: int64

Delivery Status value counts:
Delivery Status
Late delivery       98977
Advance shipping    41592
Shipping on time    32196
Name: count, dtype: int64

Late_delivery_risk value counts:
Late_delivery_risk
1    98977
0    73788
Name: count, dtype: int64

Customer Country value counts:
Customer Country
EE. UU.        106425
Puerto Rico     66340
Name: count, dtype: int64

Customer Segment value counts:
Customer Segment
Consumer       89420
Corporate      52528
Home Office    30817
Name: count, dtype: int64

Order Status value counts:
Order Status
COMPLETE           59491
PENDING_PAYMENT    39832
PROCESSING         21902
PENDING            20227
CLOSED             19616
ON_HOLD             9804
PAYMENT_REVIEW      1893
Name: count, dtype: int64

Shipping Mode value counts:
Shipping Mode
Standard Class    103153
Second Class       33806
First Class        26513
```